



Sir Syed University
of Engineering & Technology

Artificial Intelligence Lab

Project Proposal

Supervised by
Miss Fozia Khan

Artificial Intelligence Lab Project

BS Computer Science, Batch 2022F

Medical Image Analysis with MultiModal LLM

Group Members:

Name	Roll Number
Saad Ali	2022F-BCS-162
Muzamil	2022F-BCS-316
Muhammad Faisal	2022F-BCS-152

Supervisor Approval

Project Proposal

Title: Voice-Controlled Multimodal AI Medical Assistant for Image Diagnosis and Patient Interaction

Introduction:

The healthcare sector is facing increasing demand for scalable solutions that reduce patient wait times and assist in diagnostic workflows. Traditional doctor-patient interactions rely heavily on face-to-face communication and manual image assessment. Our project introduces a voice-controlled AI medical assistant capable of assessing various medical images (like MRIs, CT scans, and X-rays) and interacting with patients through natural language. This system supports both image-based diagnostics and speech-based queries using a multimodal large language model. The innovation not only offers diagnostic insights but also improves accessibility and reduces burden on medical staff.

Objectives:

- Develop a voice-interactive AI system that can interpret medical images.
- Enable patients to communicate with the system through voice inputs.
- Provide diagnostic outputs in both text and voice formats.
- Enhance the accessibility of healthcare diagnostics in underserved regions.

Success Criteria:

- Accurate diagnosis based on standard medical image datasets.
- Seamless conversion of patient voice input to text.
- High-quality text-to-speech conversion of diagnostic feedback.
- A user-friendly interface accessible through the browser.

Background and Literature Review:

The use of AI in healthcare is growing rapidly, especially in areas like medical imaging. One recent development is the integration of multimodal LLMs that can process both visual and textual information.

In their paper *“Large Language Models Encode Clinical Knowledge”* (Singhal et al., 2022), researchers demonstrated that large language models trained with medical knowledge can answer diagnostic questions and assist doctors by offering insights based on symptoms and history. Another relevant work is *“LLaVA: Large Language and Vision Assistant”* (Liu et al., 2023), which highlights how vision-language models can accurately describe images and generate meaningful text responses from them.

These research works support the idea that combining language understanding with image analysis can lead to highly capable systems, especially useful in domains like healthcare.

Expected Methodology:

Our approach will involve the following components:

- Multimodal inference via GROQ API
- Image processing to compatible formats
- Audio capture using ffmpeg & PortAudio
- Whisper model for speech recognition
- gTTS for voice feedback
- Gradio-based web UI for accessibility

Tools:

- **GROQ** – for high-speed inference
- **OpenAI Whisper** – for audio transcription
- **LLaMA 4 Maverick 17B 128E** – as the core LLM
- **gTTS** – for generating speech
- **Gradio** – for UI development

Timeline: Day 1-2:

- Setup GROQ API and LLaMA model
- Convert sample medical images to required format **Day 3:**
- Integrate audio recorder and transcription
- Set up Whisper for STT **Day 4:**
- Implement gTTS for TTS
- Connect voice outputs **Day 5-6:**
- Build interactive UI with Gradio
- Connect all components in unified workflow **Day 7:**
- Final testing, debugging, and demonstration preparation

Task Allocation:

- **Muhammad Faisal:** Image processing, LLM integration
- **Muzamil Mehmood:** Audio processing, TTS & STT integration
- **Saad Ali:** Gradio UI development, system testing

Conclusion: This proposal outlines an AI-based medical assistant that interacts with patients via voice and interprets medical images for diagnosis. The proposed system offers significant advancements in healthcare delivery by providing accessible, accurate, and efficient diagnostic support. The blend of voice, image processing, and AI models makes it a pioneering solution in multimodal healthcare technologies.

References

1. Singhal, K., Azizi, S., Tu, T., et al. (2022). *Large Language Models Encode Clinical Knowledge*. arXiv:2212.13138
2. Liu, H., Zhang, Y., Xu, Z., et al. (2023). *LLaVA: Large Language and Vision Assistant*. arXiv:2304.08485